Lesson 1.1

Undefined & Defined Terms

Point, line, plane — and the symbols that go with them

Three words we never define

Every word in geometry is built from three **undefined terms**.



Point A

a location — no size

the pin on a map

Three words we never define

Every word in geometry is built from three **undefined terms**.



Point A

a location — no size

the pin on a map



Line AB

straight, goes on forever

both ways: two arrows

Three words we never define

Every word in geometry is built from three **undefined terms**.



Point A

a location — no size

the pin on a map



Line AB

straight, goes on forever

both ways: two arrows



Plane P

flat surface, forever

a floor that never ends

Defined terms — and their symbols

Line



$\overleftrightarrow{AB}$

two arrows: goes on forever both ways

Defined terms — and their symbols

Line



$\overleftrightarrow{AB}$

two arrows: goes on forever both ways

Ray



$\overrightarrow{AB}$

starts at A — endpoint written **first**

Defined terms — and their symbols

Line



$\overleftrightarrow{AB}$

two arrows: goes on forever both ways

Ray



$\overrightarrow{AB}$

starts at A — endpoint written **first**

Segment



$\overline{AB}$

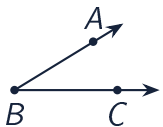
two endpoints — can be measured

Length and angle

Length a number, like 5 cm AB no bar = the **length** of $\overline{AB}$

Length and angle

Length a number, like 5 cm AB no bar = the **length** of $\overline{AB}$

Angle 

$\angle ABC$ vertex B goes in the **middle**

The mistake almost everyone makes



$\overrightarrow{AB}$ starts at A



$\overrightarrow{BA}$ starts at B

The mistake almost everyone makes



$\overrightarrow{AB}$ starts at A



$\overrightarrow{BA}$ starts at B

Watch out! $\overrightarrow{AB}$ and $\overrightarrow{BA}$ are **different** rays — they start at different points.

The mistake almost everyone makes



$\overrightarrow{AB}$ starts at A



$\overrightarrow{BA}$ starts at B

Watch out! $\overrightarrow{AB}$ and $\overrightarrow{BA}$ are **different** rays — they start at different points.

But $\overline{AB}$ and $\overline{BA}$ are the **same** segment.

Collinear, coplanar, intersection

- **Collinear**: points on the **same line**.

Collinear, coplanar, intersection

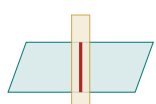
- **Collinear**: points on the **same line**.
- **Coplanar**: points on the **same plane**.

Collinear, coplanar, intersection

- **Collinear**: points on the **same line**.
- **Coplanar**: points on the **same plane**.
- Two lines **intersect** in a **point**.



Collinear, coplanar, intersection

- **Collinear**: points on the **same line**.
- **Coplanar**: points on the **same plane**.
- Two lines **intersect** in a **point**. 
- Two planes **intersect** in a **line**. 

Your turn! Pause the video and try it.

Name this figure with the right symbol.



Your turn! Pause the video and try it.

Name this figure with the right symbol.



Answer: $\overrightarrow{PQ}$ a ray — endpoint P first

Remember

- **Line** $\overleftrightarrow{AB}$: two arrows. **Ray** $\overrightarrow{AB}$: one arrow, endpoint first.
- **Segment** $\overline{AB}$: a bar. **Length** AB: no bar — a number.
- **Angle** $\angle ABC$: the vertex in the middle.

Now practice at your level:

fangmath.com → Geometry → Chapter 1